Practice Set 13 Paper 1 – Basic Sciences in Urology

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Practice paper compiled from the question bank.

Q1. Radionuclide scans and its applications in uro

Type: LAQ (25 marks) (also answered in Practice Set 2 Paper 1, Q3d–Q3f)

"Uro" is read as urology.

Principles

- Nuclear medicine shows **function, not structure**. A tracer, given in a tiny (subpharmacological) dose, follows a physiological or molecular process; the patient is the radiation source.
- **Gamma camera:** photon, collimator (admits photons from one direction), sodium iodide crystal (light), photomultiplier tubes, image.
- **PET (positron emission tomography):** the positron annihilates with an electron, giving two 511 keV photons 180° apart, detected in coincidence. This rejects noise and allows true quantitation (SUV, standardised uptake value).

	Planar	SPECT (single-photon emission computed tomography)	PET
Image	2D	3D, with CT (computed tomography)	3D, with CT or MRI (magnetic resonance imaging)
Resolution	About 20 mm	About 10 mm	About 5 mm
Quantitation	Semiquantitative	Semiquantitative	SUV
Main use	Renal function	Problem-solving	Cancer

- **^{99m}Tc (technetium-99m):** 6-hour half-life, single 140 keV photon: the workhorse. **^{18}F:** about 110 minutes, lowest positron energy, so the sharpest PET image. ^{68}Ga comes from a generator; ^{11}C (20 minutes) needs an on-site cyclotron.

Renal tracers

- Plasma clearance is about 20% filtration and 80% tubular secretion.

	^{99m}Tc-DTPA (diethylenetriaminepentaacetic acid)	**^{99m}Tc-MAG3 (mercaptoacetyltriglycine)**	**^{99m}Tc-DMSA (dimercaptosuccinic acid)**
Handling	Filtration only	Protein-bound; tubular secretion	Filtered, then retained in proximal tubules (megalin, cubilin)
Study	Dynamic	Dynamic	Static cortical at 2 hours
Use	**GFR (glomerular filtration rate)**; captopril study	**Split function, obstruction (first choice)**; transplant	**Pyelonephritis, scars**
Renal failure	Poor images	Good images	Delay imaging

Dynamic renography (MAG3)

- **Preparation:** hydrate; catheter if outlet obstruction or neurogenic bladder; cap nephrostomy. Dose 5–10 mCi IV (intravenous).
- **Perfusion phase:** 1–2 second frames; the kidney appears within seconds of the aorta.
- **Function phase: split function = area under each curve at 1–3 minutes**; normal up to 45:55. Peak at 3–5 minutes; bladder by 10–15 minutes.
- **Diuretic renography:** dilatation is not obstruction. Furosemide 0.5 mg/kg (adult maximum 40 mg) separates a patulous system from obstruction. Timing (F+20, F+15, F0, F−15) must

be stated.

T½ (half-clearance time) after furosemide	Meaning
<10 minutes	Not obstructed
10–20 minutes	Indeterminate
>20 minutes	Obstructed

Other renal applications

- **Captopril renography (DTPA):** the stenotic kidney keeps GFR by angiotensin II efferent constriction. Captopril removes this, so uptake falls. Improvement ≥10% off ACE (angiotensin-converting enzyme) inhibitors means renovascular hypertension is likely. Largely replaced by CT angiography.
- **Transplant:** acute rejection: perfusion falls and worsens; delayed graft function: perfusion kept, improves. Detects urine leak without iodinated contrast.
- **DMSA:** pyelonephritis gives an ill-defined defect that resolves by about 6 months; a scar is sharp, with cortical thinning, and persists.
- **Radionuclide cystography:** direct (tracer instilled into the bladder) or indirect (after MAG3). More sensitive and lower dose than VCUG (voiding cystourethrogram), but little anatomy. So VCUG first; nuclear cystography for follow-up.

Infection imaging

- **^{111}In-oxine leucocytes** (image at 24 hours): best for occult abdominal or genitourinary infection. ^{99m}Tc-HMPAO (hexamethylpropyleneamine oxime) leucocytes give better images, but urinary excretion obscures the tract. Neither in a transplant (leucocytes home to the graft).
- **Also know:** ^{67}Ga-citrate (non-specific, slow).

Oncology

- **FDG (^{18}F-fluorodeoxyglucose):** a glucose analogue phosphorylated and trapped (Warburg effect). Urinary excretion limits it to distant disease.
  - **Seminoma residual mass >3 cm** after chemotherapy (EAU [European Association of Urology]). Not for NSGCT (non-seminomatous germ cell tumour), because teratoma is not FDG-avid.
  - Penile cancer with palpable nodes; equivocal bladder or upper-tract metastases.
  - Poor in prostate cancer and clear cell RCC (renal cell carcinoma).
- **Prostate: PSMA (prostate-specific membrane antigen) PET** is preferred (NCCN [National Comprehensive Cancer Network]) for high-risk staging and biochemical recurrence. Specificity >90%; nodal sensitivity only 30–40%. Calais trial (2019): detection 56% (PSMA) versus 26% (fluciclovine). Bone scan (^{99m}Tc-MDP, methylene diphosphonate) remains conventional.
- **Renal mass: sestamibi SPECT** "hot" in oncocytoma; **girentuximab** (anti-CAIX, carbonic anhydrase IX) PET "hot" in clear cell RCC (ZIRCON trial).
- **Also know:** ^{123}I-MIBG (metaiodobenzylguanidine) and ^{68}Ga-DOTATATE for phaeochromocytoma; ^{99m}Tc-nanocolloid sentinel node biopsy in penile cancer; ^{99m}Tc-pertechnetate scrotal scan (torsion cold, epididymitis hot) when ultrasound is equivocal.

Therapy (theranostics)

- **^{177}Lu-PSMA-617:** beta emitter for mCRPC (metastatic castration-resistant prostate cancer) after ARPI (androgen receptor pathway inhibitor) and taxane: **VISION (2021)**. PSMA PET confirms the target first.
- **Radium-223:** alpha emitter for bone-only mCRPC.

Advantages and limitations

- Quantitative function, low radiation, safe in contrast allergy.
- Poor anatomy; PET is costly; results depend on technique.

Diagram to draw: MAG3 renogram: perfusion spike, peak at 3–5 minutes, washout; obstructed curve rising.

> **Key points to remember:**
>
> - DTPA = GFR; MAG3 = obstruction; DMSA = scars.
> - Split function at 1–3 minutes; T½ >20 minutes obstructed.
> - PSMA PET preferred; FDG for residual seminoma >3 cm.
> - Sestamibi hot = oncocytoma; ^{177}Lu-PSMA-617 (VISION).

Q2. Describe embryology and anatomy of kidney.. Describe role of nephron sparing surgery in a small renal mass.

Type: LAQ (25 marks) (also answered in 2017-Summer Paper 1, Q1 and Practice Set 1 Paper 1, Q1; NSS techniques: see Practice Set 13 Paper 2, Q1)

Embryology

- Three successive kidneys form from **intermediate mesoderm**, in cranial-to-caudal order:

Kidney	Time and site	Fate
Pronephros	Week 3–4, cervical	Non-functional; regresses
Mesonephros	Week 4, thoracolumbar	Functions briefly; tubules become efferent ductules; **mesonephric (Wolffian) duct** persists
Metanephros	Week 5, sacral	**Definitive kidney**

- **Reciprocal induction:** the **ureteric bud (UB)** grows from the caudal Wolffian duct (about week 5) into the **metanephric blastema**.
  - UB branches into ureter, pelvis, calyces and collecting ducts.
  - Blastema, induced by UB tips, forms nephrons (glomerulus to distal tubule).
- **Molecular control:** GDNF (glial cell line-derived neurotrophic factor) from the blastema acts on RET on the UB; SLIT2/ROBO2 restricts GDNF to one bud. Also WT1, PAX2.
- Urine production is established by 14 weeks; nephrogenesis ends before birth.
- **Ascent and rotation (weeks 6–9):** the kidney rises from the pelvis to the lumbar region and rotates 90° so the hilum faces medially. It takes successively higher arteries; persistent lower ones become accessory arteries.
- **Clinical correlation:** no UB gives agenesis; abnormal induction gives MCDK (multicystic dysplastic kidney); failed ascent gives ectopia; fused lower poles trapped by the IMA (inferior mesenteric artery) give horseshoe kidney; also malrotation, duplication.

Anatomy

- T12–L3; right 1–2 cm lower (liver). About 11 cm, 125–170 g. Lies on psoas: upper pole medial and posterior; angled 30–50° behind the coronal plane.
- **Relations:** right: liver, duodenum (2nd part), hepatic flexure; left: spleen, stomach, pancreatic tail, splenic flexure. Posterior: diaphragm, psoas, quadratus lumborum, subcostal nerve.
- **Coverings:** capsule, perirenal fat, **Gerota fascia** (closed above and laterally, open below), pararenal fat.
- **Structure:** cortex; 8–18 pyramids; 0.4–1.2 million nephrons.
- **Vessels (basis of partial nephrectomy):**
  - Hilum, front to back: vein, artery, pelvis.
  - **Five segmental end arteries:** posterior (first branch) plus apical, upper, middle, lower. Each can be clamped or resected separately.
  - **Brödel line:** avascular plane behind the lateral border.
  - Accessory arteries in 25–28%. **Veins anastomose freely**, so a segmental vein can be ligated.
- **Also know:** lymph drains to para-aortic (left) and paracaval or interaortocaval nodes (right).

Diagram to draw: transverse section with segmental arteries and Brödel line.

Small renal mass (SRM)

- Enhancing mass **≤4 cm (cT1a)**, mostly incidental. 20–30% are benign (about 40% at ≤2 cm). Under 2% metastasise.
- **Biology:** about 20% benign, 60% indolent, 20% potentially aggressive.
- **Work-up:** multiphase CT or MRI; chest imaging; eGFR (estimated GFR) and proteinuria (CKD [chronic kidney disease] stage); **RMB (renal mass biopsy)** when it would change management (non-diagnostic about 14%). R.E.N.A.L. nephrometry (Radius, Exophytic, Nearness to collecting system, Anterior/posterior, Location): 4–6 low, 7–9 moderate, 10–12 high complexity.

Why spare nephrons

- RN (radical nephrectomy) causes CKD, linked to cardiovascular events and death. PN (partial nephrectomy) keeps about 90% of global function versus 65–70% after RN, with equal cancer control in cT1a.

Choosing the option (AUA [American Urological Association] 2021)

Option	Role
Active surveillance	<2 cm or predominantly cystic; elderly, comorbid, life expectancy <5 years. Imaging every 3–6 months. Intervene if >3 cm or growth >5 mm a year
Thermal ablation (cryoablation, radiofrequency)	cT1a **<3 cm**, percutaneous, biopsy first. Higher local recurrence, salvaged by repeat ablation
Partial nephrectomy	**Prioritised for cT1a** needing treatment. Imperative: solitary kidney, bilateral tumours, CKD or proteinuria; also familial RCC (renal cell carcinoma), young patients
Radical nephrectomy	Only if all three: high complexity; no CKD or proteinuria; normal contralateral kidney with new GFR likely >45

- **PN principles:** negative margin (width does not matter); maximise preserved parenchyma; avoid long warm ischaemia; enucleation for familial or multifocal tumours. Approaches, steps and complications: see Practice Set 13 Paper 2, Q1.

Evidence

- **EORTC (European Organisation for Research and Treatment of Cancer) 30904 (2011):** PN versus RN for ≤5 cm: better function and similar cancer control with PN; survival slightly favoured RN, but the trial is flawed.
- **DISSRM (Delayed Intervention and Surveillance for Small Renal Masses) registry:** surveillance almost never led to metastasis; about 24% had delayed intervention.

Key points to remember:

- Ureteric bud gives collecting system; blastema gives nephrons (GDNF–RET).
- Five segmental end arteries; Brödel line; veins anastomose.
- SRM: surveillance <2 cm, ablation <3 cm, PN standard for cT1a.
- AUA: RN only if complex + no CKD + GFR likely >45.

Q3a. MOA of botulinum toxin. Indications and dosages in UROLOGY

Type: SAQ (10 marks) (also answered in 2025-Winter Paper 2, Q3d and 2017-Winter Paper 3, Q3b)

Introduction

- *Clostridium botulinum* neurotoxin, serotypes A–G. Urology uses **onabotulinumtoxinA (BoNT-A)**, the only licensed formulation.

Mechanism of action

1. The heavy chain binds the cholinergic nerve terminal; the toxin is internalised.
2. The light chain (a zinc protease) **cleaves SNAP-25** (synaptosomal-associated protein 25), so vesicles cannot fuse and **acetylcholine release stops**: chemical denervation.

- **Afferent action:** less ATP (adenosine triphosphate) and substance P; fewer purinergic (P2X3) and capsaicin (TRPV1) receptors, so urgency and pain fall.
- Lasts 6–9 months, until terminals sprout; repeat injections stay effective.

Technique

- Cystoscopic intradetrusor injection, trigone usually spared: 100 U in 10 mL at 20 sites; 200 U in 30 mL at 30 sites. Check PVR (post-void residual) at 2 weeks.

Indications and doses

Indication	Dose and evidence
Idiopathic OAB (overactive bladder), third line	**100 U** (EMBARK)
NDO (neurogenic detrusor overactivity): SCI (spinal cord injury), MS (multiple sclerosis)	**200 U** (DIGNITY); works with or without antimuscarinics
Paediatric NDO	About **10 U/kg (maximum 360 U)**, about 30 sites (Campbell series); FDA (Food and Drug Administration) label 6 U/kg, maximum 200 U
DSD (detrusor–sphincter dyssynergia)	Intrasphincteric injection, mainly men
IC/BPS (interstitial cystitis/bladder pain syndrome)	Intradetrusor, often trigonal; AUA option, off-label
BPH (benign prostatic hyperplasia)	100–300 U intraprostatic: large RCTs (randomised controlled trials) equal to saline; not used

- **Also know:** pelvic floor spasm; chronic orchialgia (RCT negative).

Complications and contraindications

- **Retention needing CISC (clean intermittent self-catheterisation)**, more in NDO; each 10 mL of baseline PVR raises the risk about 27%. The patient must accept CISC beforehand.
- UTI (urinary tract infection), haematuria; rarely generalised weakness. Avoid in active UTI, myasthenia gravis, pregnancy.

Diagram to draw: cholinergic synapse with SNAP-25 cleavage.

Key points to remember:

- Cleaves SNAP-25: efferent and afferent action.
- 100 U idiopathic OAB; 200 U NDO.
- Children 10 U/kg (maximum 360 U); FDA 6 U/kg (maximum 200 U).
- Retention needing CISC is the main risk.

Q3b. Vanishing testes

Type: SAQ (10 marks)

Definition

- A testis that **developed normally and was then lost**, usually from **torsion or a vascular accident in fetal life**, after genital masculinisation but before scrotal fixation.
- Unilateral loss gives **monorchia**; bilateral loss gives **anorchia** (testicular regression syndrome), which is rare.
- Excised "nubbins" contain haemosiderin.

Incidence

- **15–40% of nonpalpable testes** at surgery.

Clinical features

- **Contralateral compensatory hypertrophy:** volume >2 mL (sensitivity 72%, specificity 100%) or length about 19–20 mm suggests a solitary testis.
- No palpable intrascrotal "appendage" tissue; sometimes a scrotal nubbin.
- These aid counselling but **do not replace exploration**.
- **Anorchia:** micropenis in nearly half.

Evaluation

- **Imaging** (ultrasound, MRI [magnetic resonance imaging]) is not recommended: it does not change the operation.
- **Bilateral nonpalpable:** exclude DSD (difference of sex development) and CAH (congenital adrenal hyperplasia) first.
  - Anorchia: high FSH (follicle-stimulating hormone) and LH (luteinising hormone); low AMH (anti-Müllerian hormone) and inhibin B; no rise after hCG (human chorionic gonadotrophin). Most reliable during mini-puberty.

Management: diagnostic laparoscopy (after examination under anaesthesia)

Laparoscopic finding	Action
Blind-ending vessels and vas above a closed ring	Diagnosis confirmed; stop
Vessels and vas entering the internal ring	Inguinal or scrotal exploration; excise the nubbin
Testis seen	Orchidopexy (single stage or Fowler–Stephens)
Neither vas nor vessels	Search up to the kidney (agenesis very rare)

- **Excise the remnant:** viable germ cells or tubules in 5–15%.
- **Scrotal-first exploration** is reasonable with a palpable nubbin and contralateral hypertrophy (length ≥1.8 cm); laparoscopy is then needed in about a quarter.
- **Contralateral fixation:** debated; not required by the AUA (American Urological Association) algorithm.
- **Prosthesis:** offer at least 6 months after scrotal surgery or after puberty, inguinal approach.
- **Anorchia:** testosterone from puberty. **Also know:** agenesis shows ipsilateral Müllerian persistence.

Diagram to draw: laparoscopic view of blind-ending vessels at a closed ring.

Key points to remember:

- Prenatal torsion after masculinisation; 15–40% of nonpalpable testes.
- Contralateral hypertrophy (>2 mL) suggests it, but explore anyway.
- Blind-ending vessels confirm; excise remnant (germ cells 5–15%).
- Anorchia: high FSH/LH, low AMH and inhibin B.

Q3c. Management of failed pyeloplasty

Type: SAQ (10 marks)

Definition of failure

- Persistent pain, UTI (urinary tract infection) or stones; **obstructive diuretic renogram** (T½ >20 minutes); increasing hydronephrosis; falling split function.
- Dilatation settles slowly; judge by renography.
- About 1 in 9 children need a secondary procedure. Most failures occur within 1–2 years, but up to 30% occur later.

Causes

- **Ischaemia** (over-stripping) and **tension** at the anastomosis.
- **Missed crossing vessel**; inadequate excision of the narrow segment; non-dependent anastomosis.
- **Urine leak** or infection causing periureteric fibrosis.
- Missed distal stricture or high-grade VUR (vesicoureteral reflux).

Early failure (first days or weeks)

- Pain, fever, increasing dilatation: usually **oedema or clot**; temporary nephrostomy or stent, and most settle.

Evaluation of late failure

- **MAG3 (mercaptoacetyltriglycine) diuretic renogram:** obstruction and split function.
- **CT (computed tomography) angiography:** crossing vessel.
- Antegrade or retrograde pyelography (stricture length); **Whitaker test** if equivocal. Nephrostomy first if infected.

Treatment options

Option	When	Results
Endopyelotomy (retrograde or antegrade)	Short stricture, no crossing vessel; first choice after failed open repair	About 70%
Redo pyeloplasty (open, laparoscopic, robotic)	Long or dense stricture, crossing vessel	About 86–89%; best choice in most children
Ureterocalicostomy	Scarred, small intrarenal pelvis; dilated lower calyx	About 70% (redo series)
Buccal mucosa onlay (robotic, omental wrap)	Long scarred UPJ (ureteropelvic junction)	Failure about 15–25% (CORRUS)
Nephrectomy	Split function <15%, normal other kidney; multiple failures	Definitive

- **Also know:** ileal ureter or autotransplantation for long ureteric loss.

Principles of redo surgery

- Transperitoneal route in children (retroperitoneum is scarred).
- Ureteric catheter; wide mobilisation of kidney and ureter; **excise all scar**; spatulate; **tension-free, watertight** anastomosis over a stent; wrap with omentum or perirenal fat.
- **Also know:** near-infrared fluorescence shows perfusion; downward nephropexy.

Follow-up

- Stent out at 4–6 weeks; renograms for 2–3 years (late failures).

Diagram to draw: ureterocalicostomy (lower calyx to spatulated ureter).

> **Key points to remember:**
>
> - Causes: ischaemia, tension, missed crossing vessel, leak.
> - Failed open repair: endourology first; redo pyeloplasty most reliable.
> - Ureterocalicostomy for small scarred pelvis; buccal onlay emerging.
> - Nephrectomy if split function <15%.

Q3d. Circumcision undone

Type: SAQ (10 marks)

"Circumcision undone" is read as an incomplete or complicated circumcision needing revision. Foreskin restoration is mentioned briefly at the end.

Introduction

- Circumcision complications: **0.2–5%**, rising **10- to 20-fold after infancy**.
- Most late problems concern **skin quantity** (too little or too much) or the **exposed meatus**.

Inadequate ("undone") circumcision

- **Too little skin removed** or asymmetrical excision: redundant prepuce covering the glans.
- Leads to **secondary (cicatricial) phimosis**, smegma retention, balanitis, adhesions and cosmetic distress.
- A large hydrocele or hernia pushes skin forward and raises this risk.

Other complications

Complication	Management
Bleeding (commonest, about 0.1%), usually frenular	Pressure, cautery, silver nitrate or suture
Glanular adhesions; **skin bridges** (can cause chordee, torsion)	Office lysis or excision; steroids do not help
Excess skin removal, degloving	Heals by secondary intention with ointment; severe: grafts
Trapped penis (cicatrix over glans)	Betamethasone 0.05% and retraction; else relaxing incision, flaps, Z-plasty
Meatal stenosis (almost only after circumcision)	Meatotomy or meatoplasty
Subcoronal fistula; glans amputation; necrosis (cautery on clamp)	Hypospadias-type repair; reattach glans within 8 hours

Evaluation

- Original procedure (age, device); examine skin quantity, meatus, curvature; exclude hypospadias, buried or webbed penis and BXO (balanitis xerotica obliterans).

Circumcision revision

- **Indications:** redundant or asymmetrical skin, phimosis, adhesions, cosmesis.
- **Needs general anaesthesia** in children, unlike neonatal circumcision.
- **Technique (sleeve):** mark inner and outer incisions (inner 5–10 mm from the corona), excise the excess collar, haemostasis, interrupted absorbable sutures.
- Lyse adhesions and skin bridges; correct torsion or chordee; meatotomy if needed; histology if BXO suspected.

Prevention

- Separate prepuce fully and inspect the meatus before using a device (Gomco, Mogen, Plastibell); defer with hypospadias, curvature, buried or webbed penis; trained operator.

Also know: foreskin restoration

- Traction tissue expansion or skin grafting, for men wanting a foreskin; results variable.

Diagram to draw: sleeve revision showing two circumferential incisions.

> **Key points to remember:**
>
> - Complications 0.2–5%; bleeding commonest.
> - Too little skin: redundancy, phimosis, adhesions; too much: trapped penis, chordee.
> - Meatal stenosis almost only in circumcised boys.
> - Revision needs general anaesthesia.

Q3e. Female urethral diverticula

Type: SAQ (10 marks)

Definition

- **UD (urethral diverticulum):** a urine-filled periurethral sac within the periurethral fascia, connected to the urethra by an **ostium**.
- Third to seventh decades; prevalence up to 1–6%.

Aetiology

- Mostly **acquired**. **Routh theory** (most accepted): infection of **periurethral glands** leads to duct obstruction, suburethral abscess, rupture into the lumen and an epithelialised sac.
- Hence the ostium is **posterolateral at mid or distal urethra** in ≥90%.
- **Also know:** childbirth trauma theory (but 20–30% are nulliparous).

Clinical features

- "Three Ds" (dysuria, dyspareunia, post-void dribbling) in only about 5%.
- Commonest: frequency, urgency, pain, **recurrent UTI (urinary tract infection)**.
- Tender anterior vaginal wall mass; about one-third incontinent.
- **Complications:** stones (4–10%); 10% show metaplasia, dysplasia or carcinoma (**adenocarcinoma** commonest).

Evaluation

- Urine culture; **cystourethroscopy** (ostium seen in under half).
- **Imaging sensitivity: MRI (magnetic resonance imaging) 97%**, double-balloon urethrography 95%, ultrasound 82%, VCUG (voiding cystourethrogram) 74%.
- MRI shows saddlebag or circumferential extension.
- **Videourodynamics:** true SUI (stress urinary incontinence) versus dribbling from the sac.
- **Differential diagnosis:** Skene cyst (distal, distorts meatus, no communication), Gartner cyst, leiomyoma, caruncle, bulking agent.

Classification

- **L/N/S/C3 (Leach):** location, number, size, configuration, communication, continence.
- **Also know:** Leng–McGuire (absent fascia = pseudodiverticulum).

Management

- **Asymptomatic or unfit:** suppressive antibiotics, post-void stripping; counsel about cancer.
- **Symptomatic: transvaginal diverticulectomy.**
- **Principles:**
  1. Sterile urine; topical oestrogen if atrophic.
  2. **Inverted-U** anterior vaginal wall flap.
  3. **Preserve periurethral fascia** as a separate layer.
  4. Excise the **entire sac** and close the ostium.
  5. **Watertight** urethral closure over a catheter (as small as 12 Fr).
  6. **Multilayered, non-overlapping** closure; obliterate dead space.
  7. **Martius flap** if poor tissue.
- **Bothersome SUI:** concomitant **autologous pubovaginal sling**; **never synthetic mesh**.
- **Also know:** Spence–Duckett marsupialisation for distal lesions only.
- Pericatheter VCUG at 14–21 days.

Complications

- Recurrence 10–22%; de novo SUI 7–16%; urethrovaginal fistula 1–8%; stricture; recurrent UTI.

Diagram to draw: sagittal section of a mid-urethral sac and its ostium.

> **Key points to remember:**
> - Periurethral gland infection (Routh); posterolateral ostium.
> - MRI most sensitive (97%).
> - Preserve fascia; non-overlapping layers; Martius flap.
> - No synthetic mesh at diverticulectomy.

Q3f. Endoscopic procedure of SUI

Type: SAQ (10 marks)

Read as endoscopic injection of urethral bulking agents.

Introduction

- **SUI (stress urinary incontinence)** results from urethral hypermobility and **ISD (intrinsic sphincter deficiency)**.
- **Bulking agents**, injected submucosally under cystoscopic vision, are the least invasive surgical option.

Mechanism

- Bulk at the proximal urethra improves **mucosal coaptation** and resistance.
- It does not correct hypermobility, but most SUI includes some ISD.

Indications

- ISD with a **fixed, immobile urethra** (low VLPP [Valsalva leak point pressure] <60 cmH_2O, historically).
- Women wanting a **low-risk** procedure; elderly or unfit; future pregnancy.
- Persistent leakage after a sling; neurogenic SUI (safer for bladder compliance than a sling).
- **Avoid in:** active UTI (urinary tract infection), untreated detrusor overactivity.

Agents

Agent	Note
Polyacrylamide hydrogel (Bulkamid)	Non-particulate
Polydimethylsiloxane (Macroplastique)	Silicone particles
Calcium hydroxylapatite; carbon-coated zirconium beads	Particulate
Bovine collagen (skin test); PTFE (polytetrafluoroethylene); autologous fat	Withdrawn or abandoned: allergy, migration, granuloma, embolism

Technique

1. Sterile urine; local anaesthesia; cystoscope (0–30° lens) in the urethra.
2. **Transurethral:** needle into the submucosa about 1–1.5 cm distal to the bladder neck; or **periurethral:** needle alongside the urethra, watched cystoscopically.
3. Inject at 2–4 sites until the mucosal cushions **coapt in the midline**.
4. Avoid an indwelling catheter (it moulds the bulk); use CISC (clean intermittent self-catheterisation) for retention.

Outcomes

- Early improvement is common, but cure is lower than with slings and **declines with time**; repeat injections are often needed.
- **Cochrane review (Kirchin 2017):** limited evidence of benefit.
- **AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction):** an option, with counselling on durability.
- **Men:** EAU (European Association of Urology): **do not offer** for post-prostatectomy incontinence.

Complications

- Transient retention, UTI, dysuria, haematuria, urgency.
- Sterile abscess, migration, erosion; may mimic a urethral diverticulum on imaging.

Diagram to draw: cystoscopic view of coapted mucosal cushions after injection.

> **Key points to remember:**
>
> - Bulking improves coaptation; does not correct hypermobility.
> - Best for fixed urethra with ISD, or unfit patients.
> - Less durable than slings; repeat injections common.
> - Not recommended for post-prostatectomy incontinence.

Rapid Revision Sheet – Practice Set 13 Paper 1

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Radionuclide scans and its applications in uro (LAQ)

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- Ureteric bud gives collecting system; blastema gives nephrons (GDNF–RET).
- Five segmental end arteries; Brödel line; veins anastomose.
- SRM: surveillance <2 cm, ablation <3 cm, PN standard for cT1a.
- AUA: RN only if complex + no CKD + GFR likely >45.

Q3a. MOA of botulinum toxin. Indications and dosages in UROLOGY (SAQ)

- Cleaves SNAP-25: efferent and afferent action.
- 100 U idiopathic OAB; 200 U NDO.
- Children 10 U/kg (maximum 360 U); FDA 6 U/kg (maximum 200 U).
- Retention needing CISC is the main risk.

Q3b. Vanishing testes (SAQ)

- Prenatal torsion after masculinisation; 15–40% of nonpalpable testes.
- Contralateral hypertrophy (>2 mL) suggests it, but explore anyway.
- Blind-ending vessels confirm; excise remnant (germ cells 5–15%).
- Anorchia: high FSH/LH, low AMH and inhibin B.

Q3c. Management of failed pyeloplasty (SAQ)

- Causes: ischaemia, tension, missed crossing vessel, leak.
- Failed open repair: endourology first; redo pyeloplasty most reliable.
- Ureterocalicostomy for small scarred pelvis; buccal onlay emerging.
- Nephrectomy if split function <15%.

Q3d. Circumcision undone (SAQ)

- Complications 0.2–5%; bleeding commonest.
- Too little skin: redundancy, phimosis, adhesions; too much: trapped penis, chordee.
- Meatal stenosis almost only in circumcised boys.
- Revision needs general anaesthesia.

Q3e. Female urethral diverticula (SAQ)

- Periurethral gland infection (Routh); posterolateral ostium.
- MRI most sensitive (97%).
- Preserve fascia; non-overlapping layers; Martius flap.
- No synthetic mesh at diverticulectomy.

Q3f. Endoscopic procedure of SUI (SAQ)

- Bulking improves coaptation; does not correct hypermobility.
- Best for fixed urethra with ISD, or unfit patients.
- Less durable than slings; repeat injections common.
- Not recommended for post-prostatectomy incontinence.